Prediction of Historical Temperature Changes Based on Sunspots with a Neural Network

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Abstract:

Sunspots are areas on the sun where there is reduced temperature because of increased magnetic flux. These sunspots have the ability to create changes in the Earth's atmosphere's temperature. Machine learning may help predict more accurately the changes in temperature than traditional time-series prediction methods.

Based on the results, fluctuations in groundwater level located along the outer shore of Long Island may trigger groundwater "vibrational" long-term movements causing accelerated erosion underground, leading to potential landslides. With the climate showing warmer temperature fluctuations, particularly in regions with higher latitudes, the risks of landslides grow more severe, notably due to increased underground water mobility during periods previously characterized by prolonged winter cold temperatures and water mobility pauses. Therefore, understanding which areas of Long Island are prone to landslides can prevent the loss of life, damage to roads or structures, and environmental decay.

Background Information:

- Long Island's North Shore has been experiencing a series of landslides and mudslides since 2010.
- Due to climate change, there has been an increase in the severity and frequency of storms, which has led to an increase in groundwater.
- Since 1901, New York's annual precipitation has increased from 10% to 20%, and it is expected to keep increasing to 17% by the end of the century (New York State Climate Impact Assessment 2024).
- North Shore formed by headland erosion, creating steep bluffs of sandy and glacial till.
- To control further erosion, more vegetation has been planted because the roots can keep soil in place while taking in excess water in the bluffs (Fallon 2018).

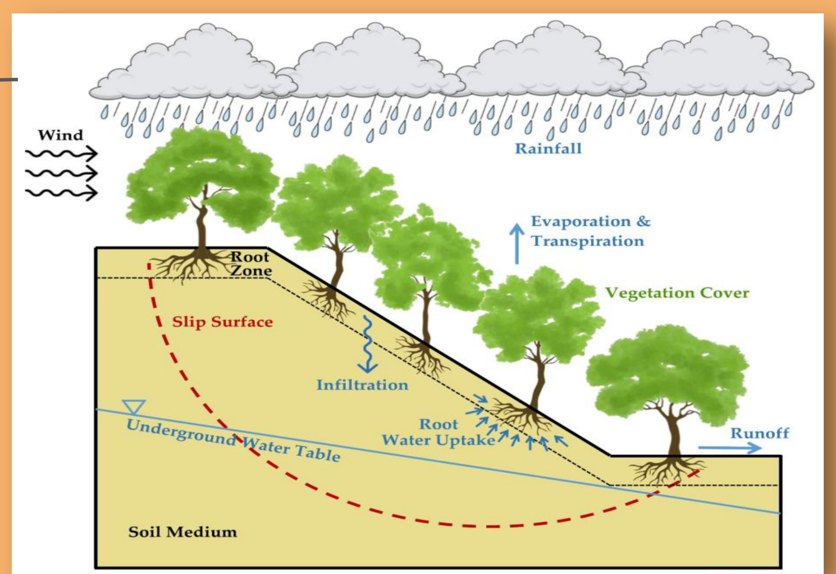

Figure 1: Conceptual figure showing the effects of fluctuating groundwater levels on landslides

Method/Materials:

- **Google Earth Pro**
  - Outlining a map of Long Island to begin looking for landslide sites
  - Pinpoint of each site and groundwater stations in close proximity
- **USGS National Water Information System**
  - Groundwater data from 11 sites
  - National Geodetic Vertical Datum of 1929
- **Global Mapper**
  - Creation of a path profile of Long Island
  - Shows sites of landslides have higher elevation
- **RStudio (v.4.2.1)**
  - Extrapolated data from NWIS
  - Fill in missing data (gaps in NWIS graphs)
  - Packages used: data.table, imputeTS, etc.
  - Moving average (kza)
  - Spectral analysis (periodograms)

Results:

$$Y_t = \frac{1}{m}\sum_{j=-k}^{k}(X_{t+j})$$

- KZ-Filter equation utilized for the moving average adapted from Zurbenko (1986); package kza (Close et al. 2020).

```
filtered ground water = groundwater - moving_average(groundwater, window(27),iterations(3))
```

- The groundwater equation is used to process raw groundwater data with a 27-day moving average filter with an iteration of three times.
- This period corresponds to the lunar cycle highlighting its potential impact on groundwater levels.

Seidman, Alyssa. Bay Avenue home in Sea Cliff is barely hanging on. Herald Community Newspapers. March 29, 2018.

Figure 2: Map of Long Island's recorded landslide occurrences (red dots) and groundwater monitoring stations by USGS (blue dots). Blue dots represent the stations used to gather groundwater data. Stations 6 and 7 are located at the same point. These sites are located in areas of higher elevation.

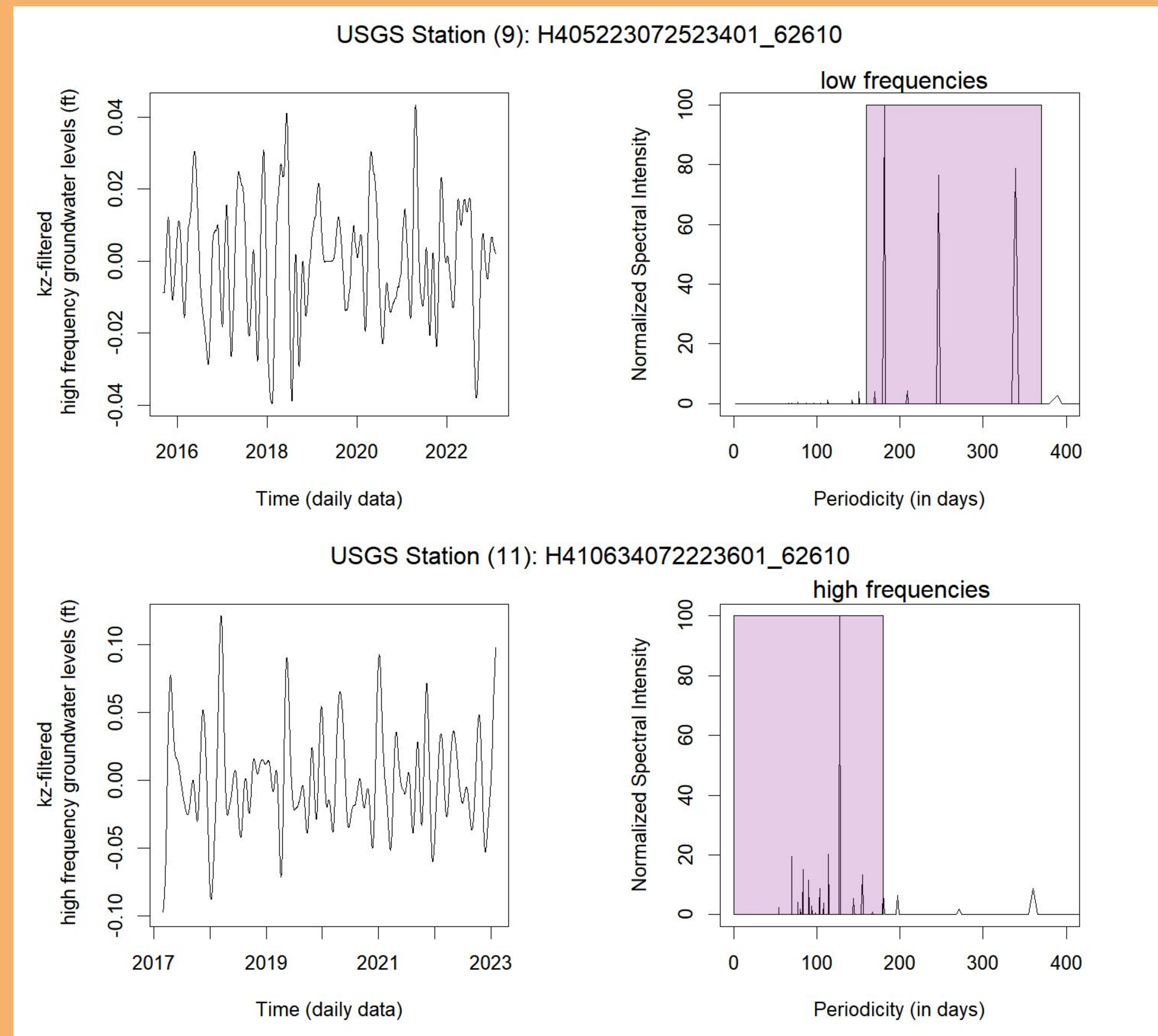

Figure 3: Groundwater fluctuations and frequency levels at USGS Sites 9 and 11.The dominant frequency peaks in spectral analysis are highlighted in purple.

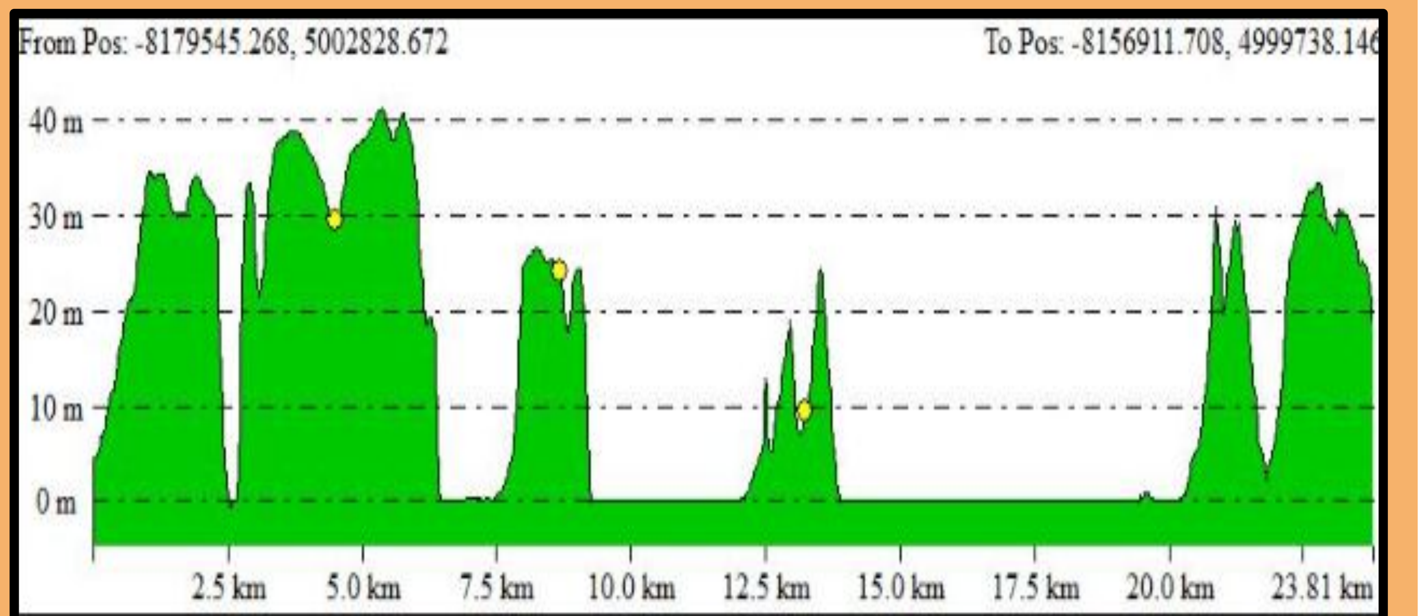

Figure 4: Path profile of Long Island's North Shore created in Global Mapper (v.24.1).

Discussion/Conclusion:

- Less than semi-annual frequencies (spikes) in groundwater —> increased risk of landslides
- GIS: high elevation —> increased likelihood of landslides
  - North shore of Long Island has steeper elevations
  - Loose sediment susceptible to erosion
- 27-day lunar cycle impacts groundwater fluctuation
- Heavy populated areas do not show annual peaks
  - Possibility it is the consumption of groundwater by those living in and around the site

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Credit Authorship Contribution Statement:

Badger, T.: editing, literature, Rstudio coding, Figure 4, Figure 5, writing- Results, Discussion; Parag, A: Literature, editing, writing- Global Mapper, Google Earth, Abstract, Results- figure 1, Methods, Discussion; Pelletier, A.: references, literature, editing, writing - Introduction, Discussion; DeRocchis, S.: figures, editing, USGS Graphs, Global Mapper, writing - Methods, Discussion; Marsellos, A.E.: supervision, Rstudio coding, guidance, editing; Tsakiri, K.G.: Rstudio coding.

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Source (Google Slides):

https://docs.google.com/presentation/d/1lNfysQl4DxyNg78qGuAE2RB−c−c−un5j8ZmaxRrXJrI/edit

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